Problem 5

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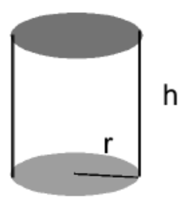
Problem 1 *Find the radius of a cylindrical container with a volume of* $2\pi\ m^3$ *that minimizes the surface area.*



(HINT : surface area , S = 2 πrh + 2 r² π;
volume , V = r² π h.)

Explanation. We have two formulas: $V = \pi r^2 h$ and $S = 2\pi rh + 2r^2\pi$. We know $V = 2\pi$ and want to find the r that minimizes S. Since we need a forumula for S only in terms of r we are going to rewrite $V = \pi r^2 h$ to give us a formula for for h in terms of r.

$$\begin{aligned}
V = 2\pi\text{and } V = \pi r^2 h \implies 2\pi &= \pi r^2 h \\
\frac{2\pi}{\pi r^2} &= h \\
\frac{2}{r^2} &= h \\
\implies S(r) &= 2\pi r\left(\frac{2}{r^2}\right) + 2r^2\pi \\
S(r) &= \frac{4\pi}{r} + 2r^2\pi
\end{aligned}$$

The domain is $(0, \infty)$.

Next, we need to minimize S.

$$\begin{aligned}
S'(r) &= \frac{-4\pi}{r^2} + 4\pi r \\
&= 4\pi\left(r - \frac{1}{r^2}\right)
\end{aligned}$$

To find the critical points:

$$
\begin{aligned}
r - \frac{1}{r^2} &= 0 \\
r &= \frac{1}{r^2} \\
r^3 &= 1 \\
r &= 1
\end{aligned}
$$

The only critical point of S is at $r = 1$. We need to verify $r = 1$ is a local extremum.

We can use the second derivative test:

$$
\begin{aligned}
S''(r) &= 8\pi r^{-3} + 4\pi \\
\implies S''(1) &= 8\pi + 4\pi > 0
\end{aligned}
$$

The function S has a local minimum at $r = 1$. So $(1, S(1)$ is the global minimum since this is the only local extremum. (See Theorem 4.5).